

These are notes for Malek Abdesselam's Functional Analysis class for Fall 2026.

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Motivation

Consider a space V of (sufficiently nice) functions $f: \mathbb{R} \rightarrow \mathbb{R}$. Then, similar to how a vector in $\mathbb{R}^n$ can be denoted as

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix},$$

we may view the function $f: \mathbb{R} \rightarrow \mathbb{R}$ as a “column vector” with uncountable index. That is, $f = (f(x))_{x \in \mathbb{R}}$. If $L: V \rightarrow V$ is a linear map, then an analogue of the finite-dimensional matrix representation can be viewed as a kernel that f is integrated against; that is,

$$(Lf)(x) = \int_{\mathbb{R}} K(x, y) f(y) dy.$$

In the special case that $L = \text{id}_V$, then the kernel $K(x, y)$ is given by the Dirac delta distribution, $K(x, y) = \delta(x - y)$. The space of these distributions is the primary object of study in Fourier analysis.

Locally Convex Topological Vector Spaces

Recall that if X is a set, then $\mathcal{T} \subseteq \mathcal{P}(\mathcal{P}(X))$ is a topology on X if it satisfies:

- $\emptyset, X \in \mathcal{T}$;
- $U_1, U_2 \in \mathcal{T}$ implies $U_1 \cap U_2 \in \mathcal{T}$;
- $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ implies $\bigcup_{i \in I} U_i \in \mathcal{T}$.

Definition: We say $(V, \mathcal{T})$ is a *topological vector space* (over $\mathbb{K}$) if V is a $\mathbb{K}$ -vector space whose topology is such that the maps

$$\begin{aligned} V \times V &\rightarrow V \\ (u, v) &\mapsto u + v \end{aligned}$$

and

$$\begin{aligned} \mathbb{K} \times V &\rightarrow V \\ (\lambda, v) &\mapsto \lambda v \end{aligned}$$

are continuous with respect to the product topology. Note that $\mathbb{K}$ is equipped with its traditional topology.

The morphisms of topological vector spaces are the *continuous* linear maps. We say $V \cong W$ are isomorphic as topological vector spaces if there is $\varphi: V \rightarrow W$ that is bijective, linear, and continuous, and φ^{-1} is also continuous.

Definition: Let V be a $\mathbb{K}$ -vector space. We say $\rho: V \rightarrow [0, \infty)$ is a *seminorm* if

$$\begin{aligned}\rho(v + w) &\leq \rho(v) + \rho(w) \\ \rho(\lambda v) &= |\lambda| \rho(v)\end{aligned}$$

for any $v, w \in V$ and $\lambda \in \mathbb{K}$.

Remark: The only difference between a seminorm and a norm is that $\rho(v) = 0$ need not imply that $v = 0$. A norm is a seminorm.

We will let $\text{SN}(V)$ be the set of all seminorms of V .

Definition: Let V be a $\mathbb{K}$ -vector space. Let $\mathcal{N} \subseteq \text{SN}(V)$. Define $\mathcal{T}(\mathcal{N})$ to be a topology with basic open sets $U(x, F, \varepsilon) \subseteq V$ given by

$$U(x, F, \varepsilon) = \{y \in V : \rho(y - x) < \varepsilon \text{ for all } \rho \in F\}$$

for any $x \in V$, $F \subseteq \mathcal{N}$ finite, and $\varepsilon > 0$.

We observe that the subsets of the form $U(x, \{\rho\}, \varepsilon)$ as ρ ranges across $\mathcal{N}$ forms a subbase for the topology of V .

Proposition: If $F \subseteq \mathcal{N}$ is finite, $\varepsilon > 0$, and $x \in V$, then

$$U(x, F, \varepsilon) \in \mathcal{T}(\mathcal{N}).$$

Proof. Suppose $y \in U(x, F, \varepsilon)$. Our goal is to find $\eta > 0$ such that $U(y, F, \eta) \subseteq U(x, F, \varepsilon)$. Suppose $z \in U(y, F, \eta)$. If $\rho \in F$, then

$$\begin{aligned}\rho(z - x) &\leq \rho(z - y) + \rho(y - x) \\ &< \eta + \rho(y - x).\end{aligned}$$

Therefore, we will define η to be small enough such that $\eta + \max_{\rho \in F} \rho(y - x) < \varepsilon$. This gives the desired result. $\square$

Notation

- The natural numbers are given by $\mathbb{N} = \{0, 1, 2, 3, \dots\}$.
- The strictly positive natural numbers are given by $\mathbb{Z}_{>0} = \{1, 2, 3, \dots\}$.
- A multi-index is a tuple $\alpha \in \mathbb{N}^n$. We let

$$|\alpha| = \alpha_1 + \dots + \alpha_n.$$

If $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, then we will write

$$x^\alpha = x_1^{\alpha_1} \dots x_n^{\alpha_n}$$

and

$$\alpha! = \alpha_1! \dots \alpha_n!$$

for the corresponding values.

If $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is any function, then we will write

$$\partial^\alpha f = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \dots \partial x_n^{\alpha_n}}.$$

- If A is a set, we will write $\mathcal{P}(A)$ for the power set of A . If A and B are sets, then the set $B^A \subseteq \mathcal{P}(A \times B)$ is the set of all functions $f: A \rightarrow B$.

- If $n \in \mathbb{N}$, then

$$[n] = \{1, 2, \dots, n\},$$

where we have $[0] = \emptyset$.

- If A is a finite set, then $|A|$ is the cardinality of A .
- Generally we will denote by $\mathbb{K}$ for either $\mathbb{R}$ or $\mathbb{C}$.